Optimization of a neural network-powered crop disease detection model for Edge Computing Devices

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1. Abstract

Plant diseases result in significant economic losses and pose an annual threat to food security in agriculture. The key to minimizing these losses lies in the accurate and prompt identification and diagnosis of plant diseases. Despite the prevalent use of deep neural networks for plant disease identification, challenges such as low accuracy and a high number of parameters persist. One agricultural application for deep learning models is the detection and classification of plant diseases through image-based crop monitoring. We used the publicly available Plant-Village dataset containing images of healthy and diseased leaves for 14 crop species and 6 groups of diseases as example data. With the modified MobileNetV3-small model, the experimental results show 98.37% accuracy during validation of system performance. The final model is in ONNX format, enabling deployment across various platforms, including mobile devices. The proposed approach achieves a favorable balance between classification accuracy and computational efficiency, making it suitable for real-time, on-device disease detection. Furthermore, the lightweight architecture reduces memory and processing requirements, enhancing its applicability in resource-constrained agricultural environments**.** These findings offer a cost-effective solution for deploying accurate deep-learning models in agricultural applications.

Keywords: Deep Learning, Plant Disease Detection, MobileNetV3-Small, Edge Computing, ONNX, Agriculture, Image Classification.

2. Introduction

Agriculture remains a cornerstone of global food security and economic stability, particularly in developing nations where a large proportion of the population depends directly on crop yield and quality for their livelihood. Among the various factors affecting agricultural productivity, plant diseases pose a persistent and critical threat, leading to substantial yield losses, reduced crop quality, and increased production costs. It is estimated that plant diseases account for significant annual losses worldwide, impacting both smallholder farmers and large-scale agricultural systems. Early and accurate detection of plant diseases is therefore essential to enable timely intervention, reduce unnecessary pesticide usage, and ensure sustainable agricultural practices. Recent advances in computer vision and deep learning (DL) have opened new avenues for automated plant disease detection using leaf images. Convolutional Neural Networks

(CNNs), in particular, have demonstrated remarkable performance in image classification tasks by automatically learning hierarchical feature representations from raw data. Numerous studies have reported high classification accuracies for plant disease detection using CNN-based architectures trained on large-scale datasets such as Plant-Village. These approaches significantly outperform traditional machine learning methods that rely on handcrafted features, thereby establishing deep learning as a dominant paradigm in agricultural disease diagnosis

Studies demonstrate the effectiveness of deep learning in plant disease classification, they often do not address the critical challenge of deploying these models on resource-constrained edge devices such as memory constraints, processing power overload and energy resources. Such high requirements limit their practical deployment in real-world agricultural environments where access to high-performance computing infrastructure and stable internet connectivity is limited, especially in rural and resource-constrained settings. Our paper tackles this challenge head-on, focusing on maximizing accuracy while operating within the resource constraints inherent to edge computing devices, thereby significantly enhancing the real-life applicability of deep learning models in agriculture.

To address these limitations, edge computing and on-device intelligence have emerged as promising solutions. Edge-based disease detection systems perform inference directly on local devices such as smartphones, embedded boards, or IoT-enabled sensors, thereby minimizing latency and reducing reliance on cloud infrastructure. However, deploying deep learning models on edge devices introduces a new set of challenges, including limited computational capacity, memory constraints, and power efficiency requirements. This necessitates the development of lightweight yet accurate models, optimized feature extraction strategies, and efficient learning frameworks that balance performance with resource utilization.

3. Related Work

Automated plant disease detection using image processing and machine learning has been an active research area over the past decade. Early approaches relied on handcrafted features combined with traditional classifiers, while recent advances have been dominated by deep learning, particularly Convolutional Neural Networks (CNNs). This section reviews existing work in three major categories: conventional image-based disease detection, deep learning–based approaches, and edge-device–oriented disease detection systems. Initial research efforts in plant disease identification primarily focused on classical image processing techniques such as color analysis, texture extraction, and shape-based features, followed by machine learning classifiers including Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Artificial Neural Networks (ANNs). These approaches required manual feature engineering and domain expertise, making them sensitive to variations in lighting, background noise, and disease severity. Moreover, their performance degraded significantly when applied to complex real-world environments, limiting their scalability and robustness for large-scale agricultural deployment. With the emergence of deep learning, CNN-based architectures became the dominant solution for plant disease classification due to their ability to automatically learn hierarchical features from raw images. Several studies demonstrated that CNNs significantly outperform traditional machine learning techniques in terms of accuracy and generalization. Pal *et al.* proposed a feature extraction framework for tomato leaf disease detection using deep learning models such as Alex-Net, VGG16, VGG19, ResNet50, and Mobile-Net, highlighting the effectiveness of CNNs in capturing discriminative disease features from leaf images **[3].** Their study emphasized the importance of reducing feature dimensionality to improve computational efficiency, especially for low-cost devices.

Similarly, Gatla *et al.* **[2]** investigated multiple pre-trained CNN architectures, including MobileNetV2, InceptionV3, ResNet50, and VGG19, for tomato leaf disease classification and reported high accuracy values exceeding 98%. Their work demonstrated that lightweight architectures such as Mobile-Net are particularly suitable for real-time disease detection while maintaining high classification performance. Other studies further explored hybrid CNN models, transfer learning, and advanced architectures to improve robustness across multiple disease classes and environmental conditions. Despite these promising results, many deep learning–based approaches rely on high computational resources and cloud-based inference, which limits their applicability in rural and resource-constrained agricultural settings. To overcome the limitations of cloud-dependent systems, recent research has shifted toward deploying deep learning models on edge devices, such as smartphones, embedded boards, and IoT platforms. Khan *et al.* proposed an optimized plant disease detection model for edge computing using

MobileNetV3-small, achieving high accuracy while significantly reducing model size through post-training quantization. **[1]** Their work highlighted the importance of balancing model accuracy with memory and computational constraints to enable real-world deployment. Similarly, Kalbande *et al.* **[4]** developed a novel edge-based plant disease detection system using MobileNetV2 deployed on a dedicated embedded platform, demonstrating reliable performance under constrained hardware conditions Their study emphasized real-time inference capability and practical usability for farmers through on-device processing without reliance on cloud infrastructure. While edge-based approaches show great promise, challenges remain in optimizing feature extraction, reducing redundant background information, and ensuring consistent performance across varying image conditions. Many existing systems still process full leaf images, which increases computational overhead and may introduce noise that affects classification accuracy.

From the reviewed literature, it is evident that deep learning has significantly advanced plant disease detection accuracy. However, a clear gap exists between high-performance models and their practical deployment on low-resource devices. There is a growing need for approaches that not only achieve high classification accuracy but also focus on efficient feature representation, reduced computational complexity, and robustness in real-world agricultural environments. Addressing these challenges motivates the development of optimized disease detection frameworks tailored for edge and low-computation platforms. In addition to model architecture design, several studies have focused on optimization strategies to reduce computational overhead while maintaining acceptable performance. Techniques such as model pruning, quantization, and knowledge distillation have been explored to compress deep learning models for deployment on resource-constrained devices. Quantization-based approaches, in particular, have shown promise by reducing model size and inference latency with minimal impact on accuracy. However, aggressive compression often leads to performance degradation, especially for fine-grained classification tasks such as plant disease detection, where subtle visual patterns play a crucial role. As a result, there is an ongoing trade-off between efficiency and accuracy that remains insufficiently addressed in many existing works.

4. Methodology

4.1 Dataset

The experiments in this study were conducted using the Plant Village dataset, a publicly available and widely used benchmark dataset for plant disease detection research. The dataset contains a large collection of high-quality leaf images captured under controlled conditions at experimental research facilities. It includes images of both healthy and diseased plant leaves, making it suitable for supervised deep learning–based classification tasks The complete Plant-Village dataset comprises 54,309 images spanning 14 different crop species, including tomato, apple, grape, potato, corn, strawberry, and bell pepper. These images are categorized into 38 distinct classes, representing healthy leaves and multiple disease types caused by fungi, bacteria, viruses, mites, and molds. Among these, the tomato subset is one of the largest and most diverse, containing images from multiple disease categories as well as healthy samples, which makes it particularly suitable for evaluating disease classification performance.

All images in the dataset are provided in RGB color format, preserving chromatic information that is critical for identifying disease symptoms such as discoloration, spots, lesions, and texture variations. The dataset is inherently imbalanced, with varying numbers of samples across different disease classes; however, its diversity in disease patterns and crop species enables robust feature learning for deep neural networks.

Due to its standardized structure, rich class diversity, and extensive use in prior studies, the Plant-Village dataset serves as a reliable benchmark for comparing model performance and evaluating the generalization capability of plant disease detection systems. In this work, the dataset was utilized to train and validate the proposed modified MobileNetV3-small model, ensuring reproducibility and comparability with existing literature.

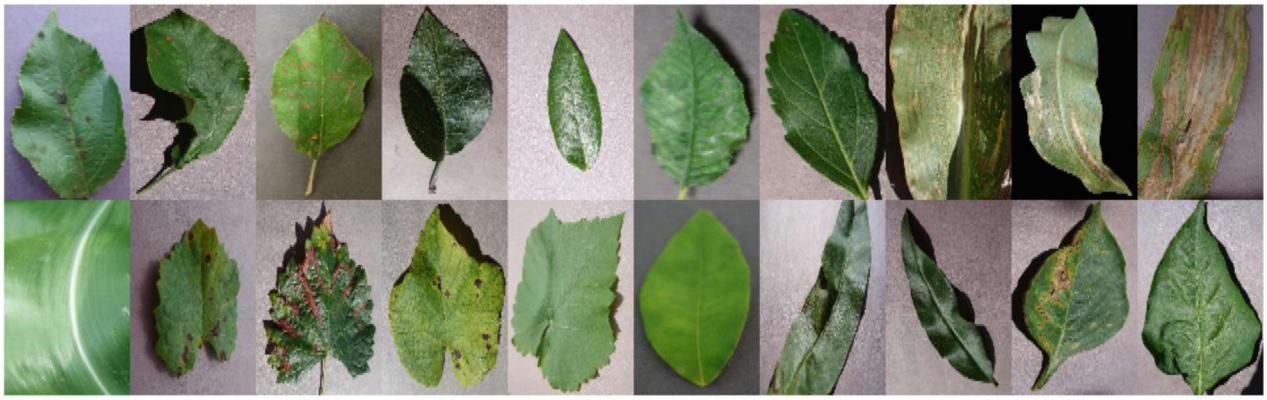

Figure 1 Sample images of the Plant-Village dataset. It is a diverse dataset with 14 plant species, including healthy and infected plants. The dataset includes a total of 54,309 image samples

4.2 Mobile Net V-3 small

MobileNetV3-Small was selected as the base architecture due to its lightweight design, low computational complexity, and suitability for edge-device deployment. In its original formulation, MobileNetV3-Small employs **Squeeze-and-Excitation (SE) attention modules** to enhance channel-wise feature representation. While SE blocks improve performance by recalibrating channel responses, they operate by globally aggregating spatial information, which can lead to the loss of fine-grained spatial cues that are critical in image-based disease detection tasks. In this work, the standard SE attention mechanism in MobileNetV3-Small was **replaced with Coordinate Attention (CA)** to improve the model's ability to capture spatially localized disease features while maintaining computational efficiency. Unlike SE attention, which performs global average pooling and focuses solely on inter-channel relationships, Coordinate Attention embeds direction-aware spatial information into channel attention by decomposing global pooling into two one-dimensional feature encoding operations along the height and width directions.

This design allows the network to preserve long-range spatial dependencies while simultaneously modeling channel interdependencies. Such spatial sensitivity is particularly important in plant disease detection, where symptoms such as lesions, discoloration, and spot patterns are often localized to specific regions of the leaf rather than uniformly distributed. By retaining positional information, Coordinate Attention enables the model to focus more effectively on disease-affected areas, improving feature discrimination.

Another motivation for replacing SE with CA is computational efficiency. Coordinate Attention introduces only a marginal increase in parameters and computational cost compared to SE blocks, making it well-suited for edge and mobile environments. At the same time, it provides superior representational capability by jointly encoding where and what information, which is essential for accurate image classification under resource constraints. As a result, the modified MobileNetV3-Small architecture achieves a better balance between accuracy, spatial awareness, and model efficiency, making it more suitable for real-time, on-device plant disease detection. This architectural enhancement contributes directly to improved validation performance while preserving the lightweight nature required for ONNX-based edge deployment.

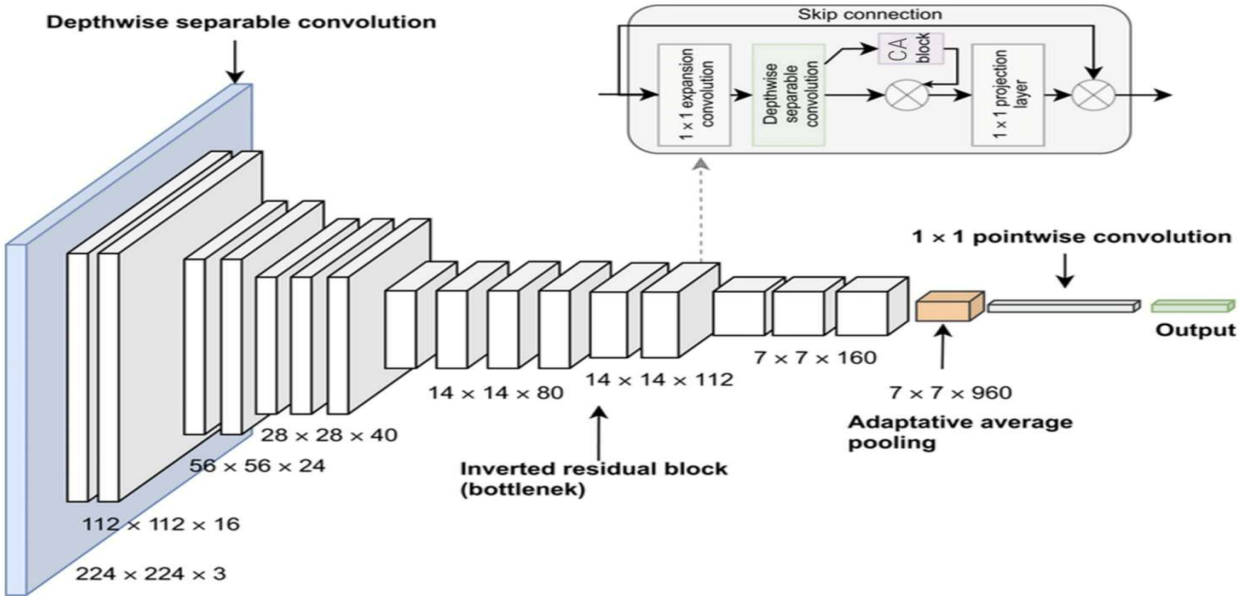

Figure 2 It shows the overall architecture of MobileNet-V3 Small. It includes a lightweight neural network design featuring depth-wise convolutions, inverted residuals, and a coordinate attention module for efficient feature extraction targeted for mobile and edge devices

4.3 Model Optimization

To ensure high classification accuracy while maintaining suitability for edge-device deployment, several model optimization strategies were employed during the training and deployment of the modified MobileNetV3-Small architecture. The model leverages transfer learning by initializing with pre-trained weights learned from large-scale image datasets, enabling faster convergence and improved generalization when applied to domain-specific agricultural data; selected layers were subsequently fine-tuned to adapt the learned representations to plant disease patterns while mitigating overfitting. MobileNetV3-Small was chosen as the backbone architecture due to its compact design based on depthwise separable convolutions and inverted residual blocks, which significantly reduce the number of trainable parameters and computational operations compared to conventional CNN architectures such as VGG and Res-Net, making it well-suited for real-time inference on resource-constrained edge devices. Furthermore, the standard Squeeze-and-Excitation (SE) attention blocks were replaced with Coordinate Attention (CA) to enhance feature representation by preserving spatial information, thereby improving localization of disease-specific patterns while introducing minimal computational overhead. To improve robustness and generalization, the input images were subjected to preprocessing and data augmentation techniques, including geometric transformations and intensity variations, which help mitigate class imbalance and reduce sensitivity to environmental variations. The training process was further optimized through careful selection of hyperparameters such as batch size and learning rate, along with the use of regularization mechanisms including batch normalization and the Hard-Swish activation function to ensure stable gradient flow and convergence. Finally, the trained and optimized model was exported to the ONNX (Open Neural Network Exchange) format, enabling cross-platform compatibility and allowing inference engines to apply graph-level optimizations such as operator fusion and constant folding, thereby ensuring efficient execution across diverse hardware platforms, including mobile and edge devices, without modifying the core model architecture.

4.4 Model Training

For model training, the MobileNetV3-Small architecture was employed using the PyTorch deep learning framework, with weights initialized from pre-training on the ImageNet dataset. Transfer learning was adopted to leverage previously learned visual representations, where the feature extraction layers were frozen and only the classification head was trained. The input images from the PlantVillage dataset were resized to a resolution of 224 × 224 pixels and fed directly to the network during training.

The training experiments were conducted on the following hardware configuration:

- Processor: AMD Ryzen™ 7 5800H with Radeon™ Graphics @ 3.20 GHz
- RAM: 16 GB
- GPU: NVIDIA RTX 3060

Although training was performed using GPU acceleration to reduce training time, the final optimized model was intended for deployment on CPU-based and edge devices.

The optimization process utilized the Adam optimizer, configured to update only the trainable parameters of the classification layer, as the feature extraction layers were frozen during training. The optimizer settings were as follows:

- Optimizer: torch.optim.Adam
- Learning Rate: 0.001
- Target Parameters: Classification head parameters only

To improve convergence and prevent stagnation during training, a learning rate scheduler was employed. The ReduceLROnPlateau scheduler dynamically adjusted the learning rate based on validation performance, with the objective of maximizing validation accuracy. The scheduler configuration was as follows:

- Scheduler: ReduceLROnPlateau
- Mode: max
- Reduction Factor: 0.1
- Patience: 7 epochs

The overall training process was governed by the following hyperparameters. A batch size of 64 was used for efficient GPU utilization, and the model was trained for a maximum of 50 epochs. The cross-entropy loss function was used for multi-class classification. Additionally, an early stopping mechanism with a patience of 20 epochs was applied to terminate training when no improvement in validation accuracy was observed, thereby preventing overfitting and unnecessary computation.

4.5 Deployment on Edge Device

After completion of model training and validation, the optimized modified MobileNetV3-Small model was exported to the ONNX (Open Neural Network Exchange) format to enable efficient deployment on edge devices such as mobile phones. ONNX provides a standardized and framework-independent representation of deep learning models, allowing seamless inference across diverse hardware platforms and runtimes without modifying the core network architecture. The ONNX-based deployment facilitates lightweight and low-latency inference, making it suitable for real-time plant disease detection in agricultural environments. By performing inference directly on mobile devices, the proposed system reduces dependence on cloud connectivity, minimizes latency, and enhances data privacy, thereby enabling practical and scalable disease monitoring solutions for farmers in resource-constrained settings.

5. Results

The performance of the proposed modified MobileNetV3-Small model was evaluated using the PlantVillage dataset, with an 80%–10%–10% split for training, validation, and testing, respectively. Model performance was primarily assessed using classification accuracy and loss trends over training epochs, as illustrated in Fig. 3 and Fig. 4.

5.1 Accuracy Analysis

Fig. 3 illustrates the training and validation accuracy curves over 50 epochs. The model demonstrates rapid convergence during the initial epochs, indicating effective transfer learning and stable optimization. Training accuracy increases sharply within the first few epochs and gradually saturates, approaching near-perfect accuracy as training progresses. Validation accuracy closely follows the training curve throughout the training process, reflecting strong generalization capability and minimal overfitting.

The proposed model achieved a peak validation accuracy of 98.37% at the 47th epoch, which represents the best-performing checkpoint during training. The narrow gap between training and validation accuracy across epochs indicates that the model effectively learns discriminative features without memorizing the training data. This behavior can be attributed to the lightweight MobileNetV3-Small architecture, the use of Coordinate Attention,

and the regularization strategies applied during training.

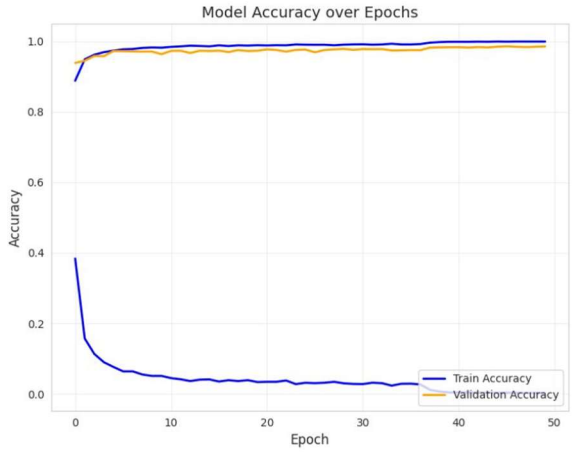

Figure 3 Training and Validation Accuracy

5.3 Loss Analysis

Fig. 4 presents the validation loss trend over the training epochs. The loss decreases significantly during the early stages of training, confirming effective gradient optimization and rapid feature adaptation. Although minor fluctuations are observed during intermediate epochs, the overall loss trend remains downward, which is characteristic of adaptive learning rate scheduling. The application of the ReduceLROnPlateau scheduler contributes to stabilizing the training process by dynamically reducing the learning rate when validation performance plateaus. In the later stages of training, a consistent reduction in validation loss is observed, reaching a minimum near the final epochs. This behavior aligns with the observed peak validation accuracy at the 47th epoch and indicates convergence toward an optimal solution.

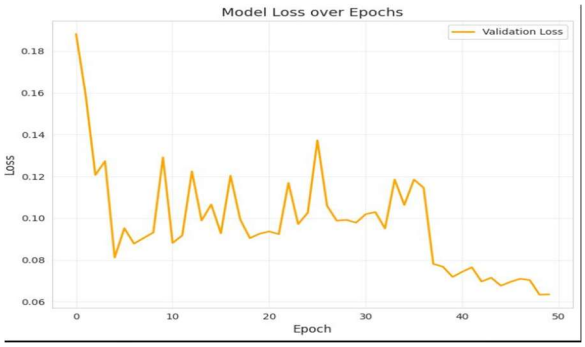

Figure 4 Loss Curve over epochs

5.4 Comparative Analysis

A comparative evaluation of the proposed model against existing state-of-the-art approaches is presented in Table 3. The results demonstrate that the proposed MobileNetV3-Small with Coordinate Attention (CA) achieves competitive performance while significantly reducing model complexity. In particular, the proposed model attains a best test accuracy of 98.37%, outperforming several existing CNN-based approaches that rely on substantially larger architectures. Compared to the AlexNet-based models reported by Mohanty *et al.* (2016), which employ approximately 60 million parameters, the proposed model achieves higher accuracy with nearly 24× fewer parameters, highlighting its superior parameter efficiency. Furthermore, the proposed approach matches the accuracy of the GoogLeNet model reported by Mohanty *et al.* (2016) (98.37%) while using only 2.5 million parameters, which is half the parameter count of GoogLeNet (5 million). This demonstrates that the integration of Coordinate Attention into a lightweight backbone can deliver competitive accuracy without the computational overhead associated with deeper architectures. Although some methods such as the modified Inception V3 model reported by Schuler *et al.* (2022) achieve marginally higher accuracy, they do so at the cost of increased model size, making them less suitable for deployment on edge devices.

Overall, the proposed model achieves a favorable balance between performance and efficiency, making it well-suited for real-time, on-device agricultural disease detection. The quantitative performance of the proposed approach can be summarized as follows:

- **Best Test Set Accuracy:** 98.37%
- **F1-score:** 0.98
- **Precision:** 0.98
- **Recall:** 0.98

These results confirm that the proposed architecture not only delivers high classification accuracy but also maintains strong class-wise consistency, reinforcing its reliability for practical deployment in mobile and edge-based agricultural setting.

Author	Architecture	Parameters	Accuracy	F1-score
Schuler *et al.* (2022)	Inception V3 (Modified)	5M	99.48%	0.9923
Proposed	MobileNet_V3_small + CA Module	2.5 M	98.37%	0.9836
Mohanty *et al.* (2016)	GoogLeNet	5M	98.37%	0.9836
Mohanty *et al.* (2016)	AlexNet	60M	97.82%	0.9782
Toda and Okura (2019)	Inception V3	5M	97.15%	0.9720
Geetharamani and Pandian (2019)	9 layers CNN	0.2M	96.46%	0.9815
Mohanty *et al.* (2016)	GoogLeNet	5M	96.21%	0.9621
Mohanty *et al.* (2016)	AlexNet	60M	94.52%	0.9449

Table 3: Results Comparison on plant village dataset

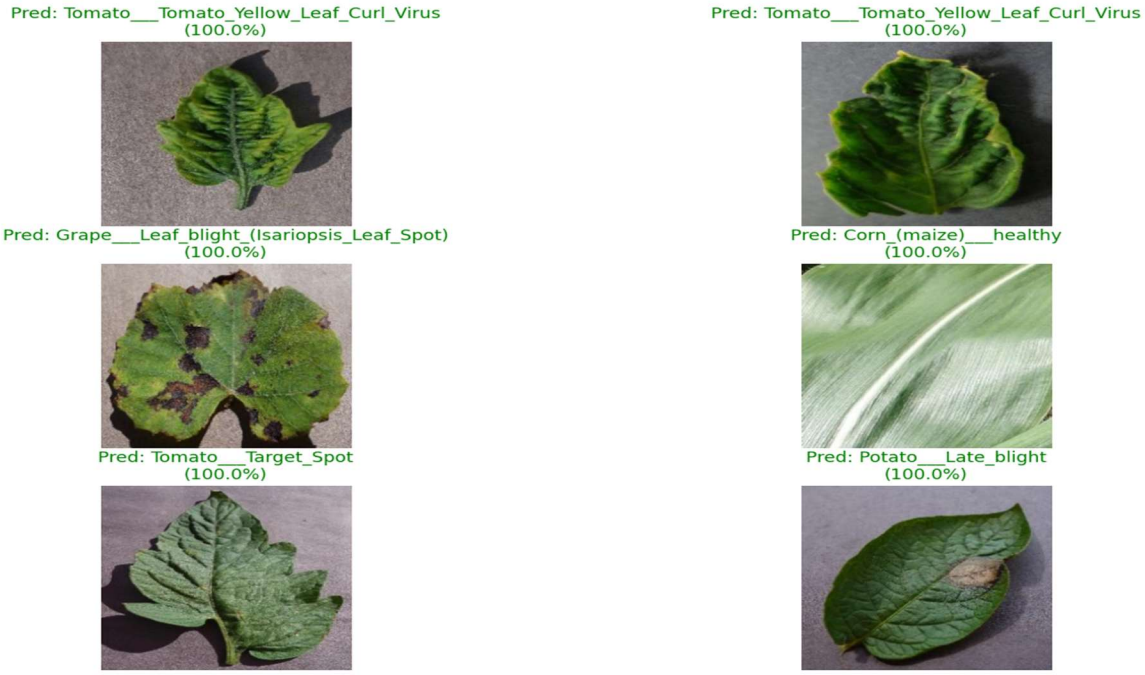

Figure 5. Sample real-world inference results obtained using the trained modified MobileNetV3-Small model, demonstrating accurate classification of healthy and diseased plant leaves across multiple crop species

6. Discussion

The experimental results demonstrate that the proposed modified MobileNetV3-Small with Coordinate Attention (CA) achieves a strong balance between classification accuracy and computational efficiency, which is a critical requirement for edge-based agricultural applications. The model attained a best test accuracy of 98.37%, along with consistently high precision, recall, and F1-score values of 0.98, indicating robust and reliable disease classification across multiple crop species. A key observation from the training and validation curves is the close alignment between training and validation accuracy, coupled with a steadily decreasing validation loss. This behavior suggests effective generalization and minimal overfitting, which can be attributed to the use of transfer learning, attention-based feature refinement, adaptive learning rate scheduling, and early stopping. The integration of Coordinate Attention plays a significant role in enhancing spatial feature representation, enabling the model to focus on localized disease patterns such as lesions, discoloration, and spots that are characteristic of plant leaf diseases. The comparative analysis further highlights the efficiency of the proposed approach. While several existing models achieve comparable or marginally higher accuracy, they do so at the cost of significantly higher parameter counts. For instance, AlexNet-based models reported by Mohanty *et al.* (2016) employ approximately 60 million parameters, yet achieve lower accuracy than the proposed model. Similarly, the proposed approach matches the accuracy of GoogLeNet-based models while using only 2.5 million parameters, which is half the parameter count of GoogLeNet. This substantial reduction in model size directly translates to lower memory usage and faster inference, making the proposed model more suitable for deployment on mobile and edge devices. The ONNX-based deployment further strengthens the practical relevance of this work by enabling framework-independent inference and facilitating deployment across heterogeneous hardware platforms. By performing inference directly on edge devices such as smartphones, the system reduces dependency on cloud infrastructure, minimizes latency, and enhances data privacy, factors that are particularly important in rural and resource-constrained agricultural settings.

Despite these promising results, certain limitations remain. The PlantVillage dataset consists of images captured under relatively controlled conditions, which may not fully represent the variability encountered in real-world field environments. Factors such as complex backgrounds, varying illumination, occlusions, and early-stage disease symptoms could impact performance in practical deployments. Addressing these challenges will be an important direction for future research.

7. Conclusion

This paper presented an optimized deep learning–based plant disease detection framework designed specifically for edge computing environments. By leveraging a modified MobileNetV3-Small architecture enhanced with Coordinate Attention, the proposed approach achieves high classification accuracy while maintaining a lightweight and computationally efficient design. Experimental evaluation on the PlantVillage dataset demonstrated a best test accuracy of 98.37%, along with high precision, recall, and F1-score values, confirming the effectiveness and reliability of the proposed model. The comparative analysis showed that the proposed model outperforms or matches several state-of-the-art approaches while using significantly fewer parameters, thereby offering a superior trade-off between performance and efficiency. The conversion of the trained model to the ONNX format enables seamless deployment on mobile and edge devices, supporting real-time, on-device disease detection without reliance on cloud connectivity.

Overall, the proposed system provides a cost-effective, scalable, and practical solution for automated plant disease monitoring in agricultural settings, particularly in resource-constrained environments. Future work will focus on evaluating the model under real-world field conditions, incorporating more diverse and noisy datasets, extending the framework to early disease detection and severity estimation, and integrating the system into end-to-end mobile or IoT-based smart farming platforms.

8. References

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